16 The two lowest frequencies at which an open narrow tube resonates are 256 Hz and 512 Hz. If one of the ends is closed, what are the two lowest frequencies at which it would now resonate?

A 128 Hz and 256 Hz

B 128 Hz and 384 Hz

C 384 Hz and 640 Hz

D 768 Hz and 1536 Hz

Length of tube $L = \frac{1}{2} \times v / 256 = v / 512$

Closed tube: $f_0 = v / (4L) = 512 / 4 = 128$ Hz

$f_1 = v / (4L/3) = \frac{3}{4} \times 512 = 384$ Hz

17 A beam of light that consists of all wavelengths between 480 nm and 600 nm is projected on a